UNIT 3 · STANDARD 6.RP.3A

Graph Ratio Tables

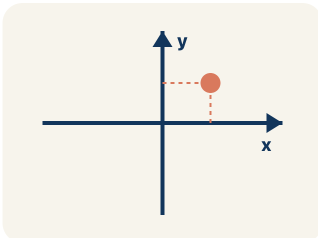
Lesson 3-3

Name: ____________________ Date: ____________________ Class: ____________________

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



A + shape made by two number lines crossing at the center point (0, 0)

Coordinate plane

Write the definition:

___

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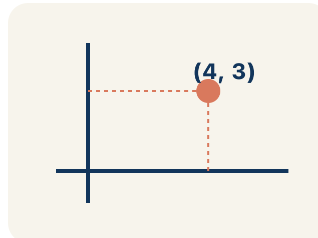

(3, 6) means go right 3, up 6

Ordered pair

Write the definition:

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(1,2), (2,4), (3,6) all line up

Linear pattern

Write the definition:

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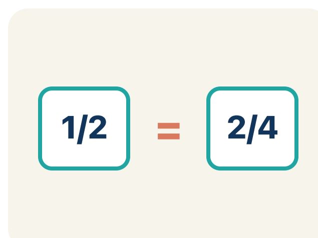


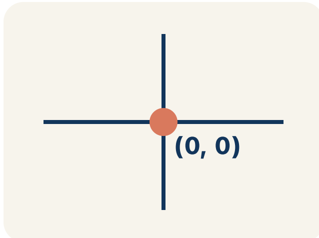
A straight line from (0,0) through (2,6) and (4,12)

Proportional

Write the definition:

___

___



The starting corner of the grid at (0, 0)

Origin

Write the definition:

Guided Notes `Level 2 Standard`

🎯 **WHAT WE'RE LEARNING TODAY**

I can graph the values from a ratio table as points on the coordinate plane.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Coordinate plane | **Ordered pair** | **Linear pattern** | **Proportional** | **Origin**

1. The flat grid formed by a horizontal and a vertical number line is the ____________ .
2. A set of two numbers (x, y) that names a point is an ____________ .
3. When graphed points form a straight line, they show a ____________ .
4. Two quantities that always keep the same ratio are ____________ .
5. The point (0, 0) where the two axes meet is the ____________ .

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

A. (8, 24)

B. (7, 20)

C. (8, 22)

D. (10, 24)

Step 1 The pattern adds 2 to x and 6 to y each time.

Step 2 After (6, 18): $x = 6+2 = 8$, $y = 18+6 = 24$.

Step 3 The next point is (8, 24).

✅ **Answer:** A. (8, 24)

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: When you graph equivalent ratios, the points will always form what shape?

A. A straight line through the origin

B. A curved line

C. A zigzag

D. A circle

Step 1 ____________________

Step 2 ____________________

Step 3 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when x = 7?

A. 28

B. 24

C. 21

D. 35

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. The smoothie recipe is 1 cup of blueberries to 2 scoops of yogurt. If cups go on the x-axis and scoops on the y-axis, which ordered pair shows 3 cups of blueberries?

A. (3, 6)

B. (6, 3)

C. (3, 5)

D. (3, 3)

Show your work:

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2. On the smoothie graph, every point follows the rule scoops = 2 × cups. Which equation describes the pattern between x (cups) and y (scoops)?

A. $y = 2x$

B. $y = x + 2$

C. $y = x \div 2$

D. $x = 2y$

Show your work:

Stretch Your Thinking
Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

Show your work:

____

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____

____

Reflect — Exit Ticket

A ratio table shows cups of rice to cups of water: (1, 2), (2, 4), (3, 6). If you plot these points, the line passes through which point?

A. (5, 10)
B. (4, 6)
C. (5, 8)
D. (6, 10)

Your answer:

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____

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Answer Key & Teacher Guide

1. **Notes 1:** Coordinate plane
2. **Notes 2:** Ordered pair
3. **Notes 3:** Linear pattern
4. **Notes 4:** Proportional
5. **Notes 5:** Origin
6. **We Try (notes):** A. A straight line through the origin *— Equivalent ratios are proportional, so their graph is always a straight line that passes through the origin (0, 0).*
7. **You Try (notes):** A. 28 *— The ratio is $y/x = 8/2 = 4$. For every x, $y = 4x$. When $x = 7$: $y = 4 \times 7 = 28$.*
8. **Try It 1:** A. (3, 6) *— Cups (x) come first, scoops (y) second. 3 cups need $3 \times 2 = 6$ scoops, so the point is (3, 6).*
9. **Try It 2:** A. $y = 2x$ *— Each cup needs 2 scoops, so y (scoops) is always 2 times x (cups): $y = 2x$. For 4 cups, $y = 2 \times 4 = 8$.*
10. **Exit Ticket:** A. (5, 10) *— The ratio is 1:2, so for 5 cups of rice you need $5 \times 2 = 10$ cups of water. The point (5, 10) continues the pattern.*